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Assessing spatiotemporal variation in drought characteristics and their dependence on timescales over Vidarbha Region, India

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ABSTRACT

In this study, a detailed spatiotemporal assessment of meteorological drought characteristics (viz., persistence, severity, and frequency) is carried out for the Vidarbha Region of Maharashtra, India, which has been infamous for the increasing number of farmer suicides there in the recent past. Using the long-term rainfall data from 11 districts for 62 years (1951–2012), the droughts in different severity classes are characterized by SPI. A detailed investigation of SPI-3, SPI-6, and SPI-12 is also done to assess the dependence of the drought characteristics at different timescales. The results reveal that the Vidarbha region is prone to meteorological droughts with frequency from once in 5 to 8 years. However, the frequency and severity of droughts have significantly increased in 1982–2012 compared to 1951–1981. The persistence of droughts is well captured by SPI at longer timescales. The trend analysis reveals the drought severities to be increasing over 10 out of 11 districts.

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Climate change; drought trend; meteorological drought; SPI; Vidarbha Region

1. Introduction

Drought is one of the most catastrophic natural phenomena mankind has been battling ever since its evolution. The remarkable fact about droughts is that they prevail for longer duration and affect large areas (or regions) as well as population compared to other kinds of natural calamities (Mirabbasi et al. 2013; Mishra and Singh 2011; Rossi et al. 1992; Wilhite 2007; Wilhite and Glantz 1985). Droughts are broadly classified into meteorological, agricultural, hydrological and socio-economic types. A meteorological drought is described as a long-term deficit of rainfall compared to the normal conditions, which transmits to other types of droughts. Meteorological droughts are thus defined over long/extended temporal scales, such as multi-monthly, seasonal, or annual periods (Ghozat et al. 2022; Hosseini et al. 2020; Sharafati et al. 2020; Swain et al. 2020, 2021; Wilhite et al. 2007, 2014). Drought indices are commonly used to identify the drought events and their characteristics. These indices simplify the complex linkages of the hydroclimatic variables, thereby making it convenient to quantify the intensity, frequency, duration, spatial

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coverage of the climatic anomalies, i.e., hydrologic extremes (Swain et al. 2022a, 2022b). Moreover, they can effectively capture the processes, such as start, end and continuance of droughts that are vital aspects for the risk assessment (Amrit et al. 2018a, 2018b; Nabaei et al. 2019). Therefore, these indices are important tools to assess and monitor the droughts (Mishra et al. 2015; Montaseri and Amirataee 2017; Swain et al. 2021).

Several meteorological drought indices exist in literature and each of them is associated with specific advantages and limitations. However, Standardized Precipitation Index (SPI) developed by McKee et al. (1993) is the most widely used index for drought monitoring (Alamgir et al. 2015; Almedeij 2014; Dabanlı et al. 2017; Hao et al. 2017; Lloyd-Hughes and Saunders 2002; Yaseen et al. 2021). As the computation of SPI is simple and based on the probability of precipitation for any timescale, it can be used for comparing drought conditions over diverse climates (Bordi et al. 2004; Guttman 1998; Heim 2002; Keyantash and Dracup 2002; Rouault and Richard 2005; Svoboda and Fuchs 2016; Svoboda et al. 2012; Thomas et al. 2015; Tsakiris and Vangelis 2004; Vicente-Serrano 2006; Wang et al. 2015). Moreover, the SPI-based transition of near normal to moderate, moderate to severe, and severe to extreme conditions is uniform for both dry and wet conditions (Guttman 1999; Heim et al. 2017; Paulo and Pereira 2006). The "Lincoln Declaration on Drought Indices" (Hayes et al. 2011) recommended the National hydrological and meteorological services all over the world to use SPI as the meteorological drought indicator. It also enunciated that the World Meteorological Organization (WMO) should foster the extensive application of SPI to monitor meteorological droughts worldwide. Further, several researchers (Halwatura et al. 2017; Hayes 2006; Keyantash and Dracup 2002; Khatiwada and Pandey 2019; Montaseri and Amirataee 2017; Morid et al. 2006; Ntale and Gan 2003; Quiring 2009) have evaluated meteorological drought indices across different parts of the world and have found SPI to be superior.

The inherent benefit of SPI is its usability in different timescales. Due to this, there are many studies related to SPI application in multiple timescales (Angelidis et al. 2012; Bazrafshan et al. 2014; Dogan et al. 2012; Jain et al. 2010; Kattelus et al. 2016; Logan et al. 2010; Mukherjee et al. 2018; Potop et al. 2012; Rouault and Richard 2005; Shukla et al. 2011; Svoboda et al. 2015; Vicente-Serrano 2006). Among them, moving windows of rainfall at 3-monthly, 6-monthly and 12-monthly timescales are most commonly found in drought literature (Bazrafshan et al. 2014). In general, SPI based on 3-months account for short-term droughts. In other words, it can be used to capture the seasonal droughts whereas SPI-6 and SPI-12 are used for representation of intermediary and long-term droughts, respectively (Angelidis et al. 2012; Logan et al. 2010; Rouault and Richard 2005; Vicente-Serrano 2006). The purpose of SPI-based multi-scalar assessment is to study the dependence of drought characteristics (viz., persistence, severity, and frequency) on the timescales used.

In addition to the aforementioned drought characteristics, trend is also a crucial factor, which accounts for the presence of monotonicity in the time series. The quantification of drought trends helps decision-makers to plan and manage the water resources efficiently, thereby reducing the dreadful impacts of droughts. The non-parametric Mann-Kendall (MK) test is the most widely used method for trend analysis of hydro-meteorological variables. The monotonicity in drought characteristics is investigated by applying MK or Modified Mann-Kendall (MMK) tests to SPI series in several prior studies (Botai et al. 2016; Chen et al. 2009; Ganguli and Reddy 2014; Khadr 2017; Mahfouz et al. 2016; Mallya et al. 2016; Naresh Kumar et al. 2012; Omonijo and Okogbue 2014; Sheffield and Wood 2008; Thomas et al. 2015). It is observed in several of these studies that the trend of droughts varies with timescale; hence, discerning a clear picture on an actual trend of

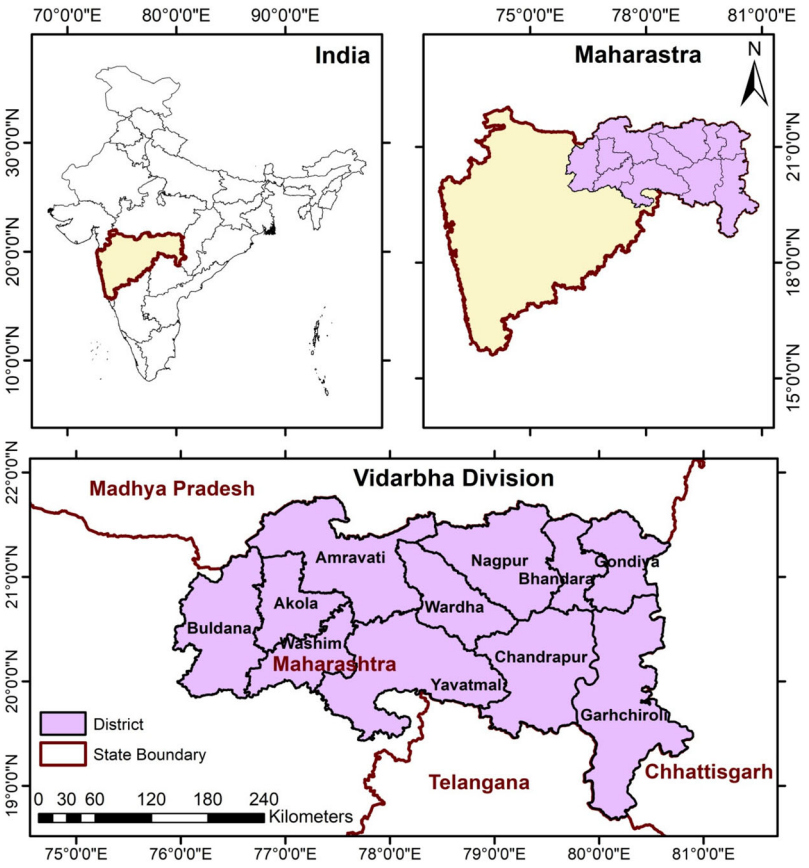


Figure 1. Location of the Vidarbha Division, Maharashtra, India.

droughts is a difficult endeavor. The SPI series may possess differences in terms of randomness at multiple timescales; however, to best of the author's awareness, no previous study has explicitly addressed the cause of such inconsistencies in drought trends.

Considering the above, this study has been carried out to address the following: 1. A detailed spatiotemporal assessment of meteorological droughts characterized by SPI using long-term rainfall records. 2. Frequency analysis of droughts in total and in different severity categories for entire duration as well as in epochs to understand the climate change signals. 3. Multi-scalar assessment of droughts using SPI at different moving timescales and impacts on drought characteristics. 4. Detailed investigation of trend in droughts and its relevance with SPI timescales.

2. Study area and descriptive statistics of the data

In this research, the Vidarbha Region of Maharashtra, India, is taken as the study area for the assessment of meteorological droughts. It lies in the northeastern part of the state. The spatial extent is from 18.63° N to 21.88° N latitudes and 75.85° E to 80.90° E longitudes, covering an area of 97321 km,^{2} which is nearly one-third area of the state. The Vidarbha division is bordered on the north by Madhya Pradesh, on the east by Chhattisgarh, and on the south by Telangana, whereas Khandesh and Marathwada divisions of Maharashtra lies to its west (Figure 1). The Vidarbha division has a population

Table 1. Statistical properties of annual rainfall in different districts over Vidarbha division.

		Annual rainfall statistics					
S. No.	Districts	μ	σ	Max.	Min.	C_v	S.E.
1	Yevatmal	979.6	223.9	1551.4	570.0	0.23	28.21
2	Washim	883.0	219.1	1441.2	412.3	0.25	27.60
3	Wardha	1001.5	227.9	1473.9	336.2	0.23	28.72
4	Nagpur	1092.4	210.7	1616.0	623.3	0.19	26.54
5	Gondia	1361.2	312.9	2463.4	828.3	0.23	39.43
6	Gadchiroli	1428.5	331.7	2399.4	903.9	0.23	41.79
7	Chandrapur	1281.4	303.1	2238.4	761.1	0.24	38.18
8	Buldana	750.8	168.8	1145.0	449.4	0.22	21.26
9	Bhandara	1265.1	260.6	2003.9	771.1	0.21	32.83
10	Amravati	913.8	200.4	1366.5	527.2	0.22	25.25
11	Akola	787.9	203.0	1334.9	449.2	0.26	25.58

Note: μ = Mean, σ = Standard deviation, Max. = Maximum, Min. = Minimum, C_v = Coefficient of variation, S.E. = Standard error. Here, μ, σ, Max., and Min. are expressed in millimeters (mm).

of 23 million as per the 2011 Census of India. Under this division, there are eleven constituent districts, i.e., Yevatmal, Wardha, Washim, Chandrapur, Gadchiroli, Gondia, Bhandara, Buldana, Akola, Amravati, and Nagpur. The land use of the region is mostly dominated by agriculture, which is also the primary source of income for majority of the population (Thakre and Thakre 2016). The average annual rainfall over the region is 1100 mm, most of which is concentrated to the monsoon season (mid-June to mid-October). Droughts have been common in the division in recent years. In less than a decade, more than 20,000 farmer suicides in Maharashtra are reported, with 70% of them occurring in the Vidarbha division, owing to a bleak financial situation exacerbated by crop failures due to recurrent droughts. Hence, a comprehensive spatiotemporal assessment of droughts across the region is required, which will be helpful in developing appropriate planning and management measures.

The monthly rainfall data for all eleven districts were collected for a period of 62 years (1951–2012) from India Meteorological Department (IMD). The mean annual rainfall possesses a remarkable variation across different districts, i.e., 751 mm in Buldana to 1428 mm in Gadchiroli. Various statistical properties, viz., mean, standard deviation, maximum value, minimum value, coefficient of variation and standard error of annual rainfall for different districts are presented in Table 1. From maximum and minimum annual rainfall values of different districts, it can be inferred that the rainfall possesses significant inter-annual variability in the region. This is also reflected from the coefficient of variation, which is high for all districts (lowest value = 0.193), inferring the annual rainfall to be frequently anomalous. It is pertinent to mention that the observed rainfall was available up to 2012; however, it is believed that the duration of 62 years is reasonably long to identify the drought characteristics of the districts with substantial confidence, and the objectives of the study can be fulfilled efficiently.

3. Methodology

3.1. Framework of the study

In this study, SPI is used for characterization/analysis of droughts in Vidarbha division. The overall framework of methodology is presented in Figure 2. As seen, in the first part of the study, the annual rainfall time series for 1951–2012 is prepared from monthly rainfall available for all eleven districts. Using the derived SPI-Annual, the total number of droughts (SPI< -1) having occurred in each district during 1951–2012 is estimated, and

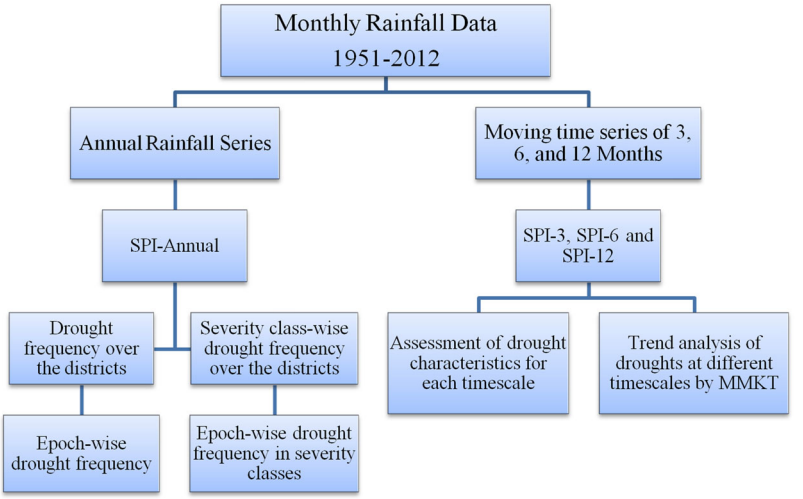


Figure 2. Flowchart depicting the overall methodology of the study.

thus, frequency of drought occurrences in each district derived. Based on SPI-values, the total number of droughts are further divided into three severity categories (moderate for $-1.5 < \text{SPI} < -1$, severe for $-2 < \text{SPI} < -1.5$, and extreme for $\text{SPI} < -2$), and frequency in each severity category analyzed. The drought characteristics get worsened under the influence of climate change, as reported in several recent studies (Aadhar and Mishra 2020; Mishra et al. 2020; Qutbudin et al. 2019; Tabari and Willems 2018). Further, Mishra and Singh (2010) found that India had been prone to frequent droughts during post-1980 period. It is for this reason that the total duration of 62 years has been divided into two epochs of 31 years, 1951–1981 and 1982–2012, in this study, and frequency is analyzed for both the epochs. This will help assessing if there is any change in drought characteristics in recent decades. The number of droughts in total and in each severity class is analyzed for both the epochs.

In the other part, the long-term monthly rainfall records for 1951–2012 are converted to time series of moving accumulations of 3-months, 6-months, and 12-months. These timescales are most commonly used in drought literature. For each moving timescale, the corresponding SPI time series (designated as SPI-3, SPI-6 and SPI-12, respectively) is computed. The drought characteristics, viz., persistence, severity, and intensity of droughts at each timescale is assessed for different districts using the Run Theory (Yevjevich 1967). Further, the trend analysis of droughts in different districts is carried out using the non-parametric Modified Mann-Kendall test (MMKT) at different timescales, and results are analyzed. Notably, MMKT is an improved version of the conventional Mann-Kendall test for it accounts for autocorrelations among the data (Hamed and Rao 1998), and therefore, is superior in terms of accuracy. A detailed description of SPI, Run Theory and MMKT is provided in the following sub-sections.

3.2. Standardized precipitation index

SPI is computed by fitting an appropriate probability distribution (preferably 2-parameter Gamma) to long-term rainfall records and then transforming the cumulative distribution function (CDF) to a standard normal distribution (mean $= 0$, standard deviation $= 1$). Thus, 99.8% of the values are in the range of -3 to $+3$. The positive (negative) values of

Table 2. Categorization of SPI values into different conditions (McKee et al. 1993).

Class	SPI
Extremely Wet	≥ 2
Very wet	1.5 to 1.99
Moderately wet	1.0 to 1.49
Normal	−0.99 to 0.99
Moderate drought	−1.0 to −1.49
Severe drought	−1.5 to −1.99
Extreme drought	≤ -2

SPI indicate higher (lower) precipitation than the median. The normal condition is represented by SPI values between −1 to 1. When the SPI falls below −1, it is regarded as a drought situation. The detailed computation procedure of SPI (i.e., by fitting Gamma distribution to precipitation time series and then transforming the CDF to a standard normal variate) can be referred from Swain et al. (2021). SPI can be computed using *standard-precip* package in the Python programming language.

As SPI is derived through the equiprobability transformation, it has the advantage of considering the probability of precipitation over several time scales, allowing it to anticipate short-, intermediate-, and long-term droughts. Further, SPI can represent both dry and wet conditions uniformly (Bazrafshan et al. 2014). The categorization of SPI values into different dryness/wetness conditions is presented in Table 2. It is evident that a drought (SPI ≤ -1) can be recognized in moderate, severe or extreme class based on the severity.

3.3. Run theory

The 'Run Theory' was proposed by Yevjevich (1967) to identify the drought events and examine their properties, i.e., initiation, end, length/duration, severity and intensity. A schematic representation of the theory is presented in Figure 3. The threshold level (X_0) is the most basic factor for recognizing these properties. A run is a segment of the time series of a drought variable (X_t), where all values are either above or below the threshold, thereby being regarded as either a positive or negative run. In this study, SPI is the drought variable X_t. The threshold X_0 is set at SPI of −1, below which, it is a negative run indicating a drought event. For a particular negative run (or a drought event), the time X_t falls below X_0 is the initiation of drought (t_i). In the subsequent time steps, when X_t goes above X_0, it is regarded as the end of the run (t_e). The time between t_i and t_e of a negative run is considered as the drought duration (D_d), which indicates the persistence of a drought event. The cumulative sum of SPI values during a drought event is regarded as the drought severity (S_d), whereas the drought intensity (I_d) is the average value of SPI during a drought run. Therefore, for a particular drought event, I_d can be computed as the ratio of S_d and D_d (Mishra and Singh 2010).

3.4. Modified Mann-Kendall test

A positive (negative) value of Z_{MMK} signifies an increasing (decreasing) trend. If the magnitude of Z_{MMK} is considered significant at 1%, 5% and 10% if its magnitude is greater than 2.576, 1.960 and 1.645, respectively (Sahoo et al. 2021; Swain et al. 2019, 2022; Tiwari and Pandey 2019). Z_{MMK} is computed as,

$$Z_{MMK} = \begin{cases} \dfrac{S-1}{\sqrt{V^*(S)}} & if\ S > 0\ 0\ if\ S = 0 \\ \dfrac{S+1}{\sqrt{V^*(S)}} & if\ S < 0 \end{cases} \quad (1)$$

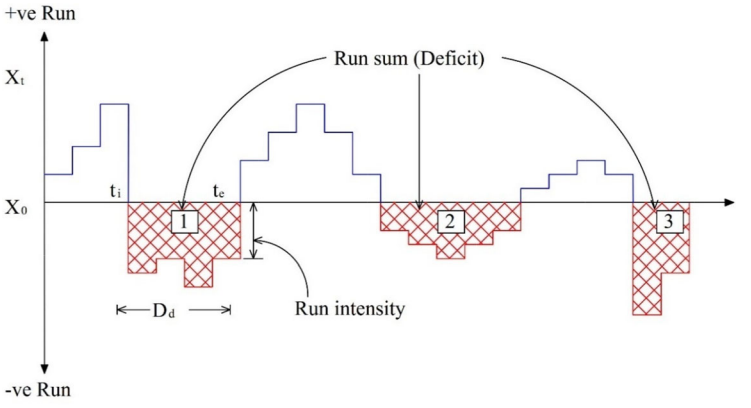


Figure 3. Run theory for estimation of drought characteristics.

$$\text{where} \quad S = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} sgn(x_j - x_i)$$

Here, x_i and x_j are the data values of a time series in sequence, n is the total number of data values, sgn refers to the signum function such that,

$$sgn\ (x_j - x_i) = \{+1\ if\ (x_j - x_i) > 0\ \ 0\ if\ (x_j - x_i) = 0\ \ -1\ if\ (x_j - x_i) < 0 \qquad (2)$$

The variance of the MMKT i.e., V*(S) can be computed as,

$$V^*(S) = VAR(S) \cdot C_f \qquad (3)$$

Here, *VAR(S)* is the variance of the conventional MK test, which is modified by a correction factor (C_f) to account the effects of autocorrelation among data points.

$$VAR(S) = \frac{n(n-1)(2n+5) - \sum_{i=1}^{m} t_i(t_i-1)(2t_i+5)}{18} \qquad (4)$$

where, t_i is considered as the number of ties up to sample i.

$$C_f = 1 + \frac{2}{n\ (n-1)(n-2)} \times \sum_{i=1}^{n-1} (n-i)(n-i-1)(n-i-2)\rho_s(i) \qquad (5)$$

where, $\rho_s(i)$ is the serial autocorrelation in the series (Swain et al. 2021). The MMK test can be carried out using *pymannkendall* package in the Python programming language.

4. Results and discussion

4.1. Drought frequency analysis using SPI-annual

The drought characteristics for each district are identified from the SPI computed using the annual rainfall data for 62 years (1951–2012). When SPI of a particular district is less than −1, it is regarded as a drought-affected year or an annual drought event. The district-wise distribution of the number of drought-affected years is presented in Table 3 and depicted in Figure 4. It can be observed that the number of annual drought events in the

Table 3. The details of drought frequency in different severity classes for all the districts over Vidarbha division (SPI).

Phase	1951–1981				1982–2012				1951–2012			
Stations	Moderate	Severe	Extreme	Total	Moderate	Severe	Extreme	Total	Moderate	Severe	Extreme	Total
Yevatmal	3	1	0	4	3	2	1	6	6	3	1	10
Washim	2	1	1	4	5	0	1	6	7	1	2	10
Wardha	3	2	0	5	5	0	1	6	8	2	1	11
Nagpur	1	2	1	4	5	2	0	7	6	4	1	11
Gondia	2	1	0	3	2	3	0	5	4	4	0	8
Gadchiroli	3	1	0	4	5	2	0	7	8	3	0	11
Chandrapur	2	1	0	3	5	3	0	8	7	4	0	11
Buldana	3	1	0	4	4	3	0	7	7	4	0	11
Bhandara	2	1	0	3	3	1	1	5	5	2	1	8
Amravati	2	2	0	4	3	3	0	6	5	5	0	10
Akola	2	1	0	3	4	2	0	6	6	3	0	9

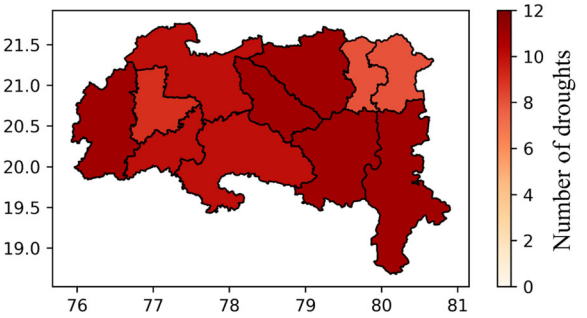


Figure 4. Total number of droughts in Vidarbha division in 1951–2012.

Vidarbha region has varied from 8 to 11 during the period of 62 years (Table 3), implying that the frequency of droughts has generally been once in 5 to 8 years.

However, this estimation of frequency does not give an account of the severity of the drought events. Therefore, based on the SPI values, droughts are categorized into moderate, severe and extreme severity classes. Figure 5 presents the spatial variation of frequency of droughts in each severity class in the Vidarbha region during 1951–2012. It can be seen that most districts have undergone high number of moderate droughts compared to severe or extreme severity classes. The frequency of moderate droughts varied from 4 to 8 in different districts whereas the frequency of severe droughts varied from 3 to 5. Only two districts (i.e., Gondia and Amravati) witnessed equal number of moderate and severe droughts (Table 3). Overall, these districts have undergone high number of severe droughts. Further, 5 districts (Yevatmal, Washim, Wardha, Nagpur and Bhandara) have at least faced once an extreme drought. Washim is the only district to have faced extreme droughts twice in 62 years and higher number of extreme droughts than that of severe droughts (Table 3).

As also mentioned earlier, the drought characteristics get further worsened due to looming climate change (Kumar et al. 2013; Mishra 2020; Nam et al. 2015). Several studies (Aadhar and Mishra 2020; Gosain et al. 2006; Mishra et al. 2020; Tabari and Willems 2018; Van Loon et al. 2014) have projected the frequency of hydro-meteorological extremes to increase in future due to higher rainfall aberrations. Further, India has been prone to frequent droughts during post-1980 period (Mishra and Singh 2010). Hence, the frequency of droughts is analyzed epoch-wise, i.e., for 1951–1981 and 1982–2012. The total number of droughts during 1951–1981 and 1982–2012 are presented in Figure 6. It

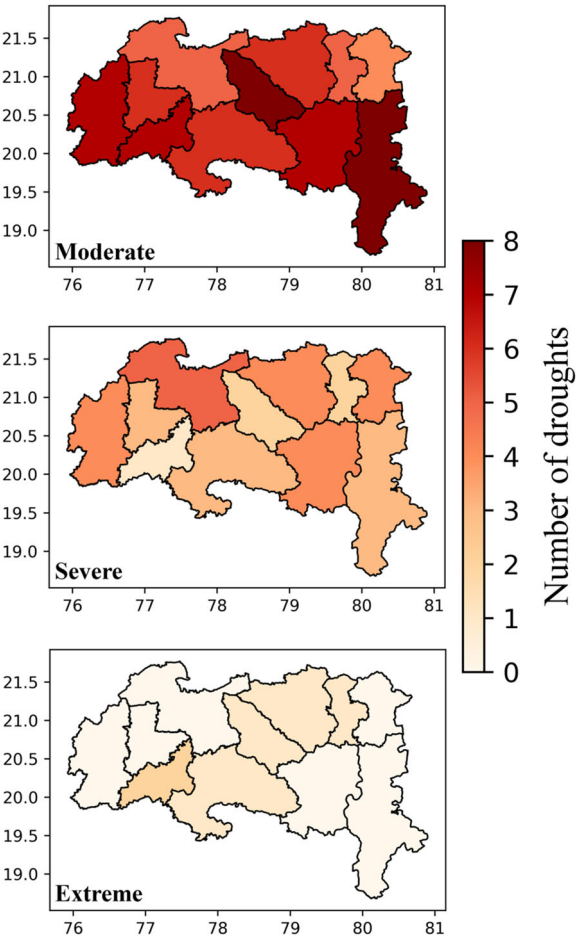


Figure 5. Number of droughts in Vidarbha division in 1951–2012 in different severity classes.

can be observed that the number of droughts has increased significantly in the latter period compared to the former in all districts of the Vidarbha division. The frequency in different districts varied from 3 to 5 during 1951–1981 in contrast to 5 to 8 during 1982–2012, which implies that the droughts have occurred with higher frequency in different districts in the latter part. The increase in frequency has been minimum in Wardha district; five in the first and six in the second epoch. On the other hand, the maximum increase in frequency has been observed in Chandrapur district; three and eight in the first and second epochs, respectively. It can thus be seen from Table 3 that the total number of droughts in most districts has increased nearly two-fold in the latter epoch.

A comparative representation of the spatial variation of drought frequency in three different severity classes for both the epochs are shown in Figure 7. The drought frequencies in total and in different severity classes for the entire duration as well as for epochs are summarized in Table 3. The number of moderate droughts in 1982–2012 over all the districts is higher than that of 1951–1981 except two districts (Yevatmal and Gondia), where it is equal in both the epochs. The number of severe droughts has also increased (i.e., doubled/tripled) for the latter epoch over most of the districts as compared to the former. Further, the number of districts to have experienced an extreme drought during the first epoch is two, which rose to four for the second epoch. Therefore, the droughts have not

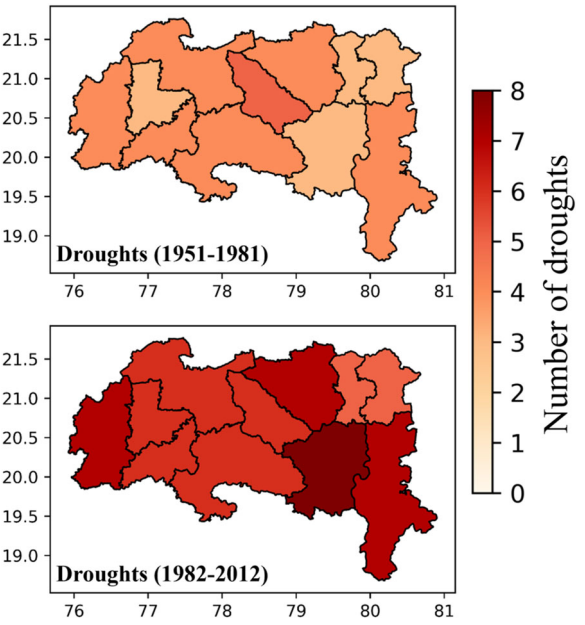


Figure 6. Total number of droughts in Vidarbha division during 1951–1981 and 1981–2012.

just increased in terms of frequency; they have also increased in terms of severity. Since the Vidarbha division is mainly covered by agricultural lands, such triggered drought events may lead to serious repercussions.

4.2. Dependence of drought characteristics on timescale

The moving accumulation of 3-, 6- and 12-month rainfall is used to compute SPI to assess the dependence of drought characteristics on timescales. Generally, the SPI based on 3-months accounts for short-term droughts, and thus, can be used to capture seasonal droughts whereas SPI-6 and SPI-12 represent intermediary- and long-term droughts, respectively (Djellouli et al. 2016; Logan et al. 2010; Vicente-Serrano 2006). SPI at different timescales is computed for all the districts of Vidarbha, and drought attributes determined using the Run Theory. The number of drought events and total duration (months) of droughts at all three timescales for the districts of Vidarbha Region is shown through radar charts in Figure 8. The exact values are also presented in Table 4. It can be observed that the total number of drought durations for all districts at different time scales is similar, but the number of events reduces with increase in time scale from 3-months to 12-months. Only the Washim district is an exception, where the number of drought events at 3-monthly scale is lower than that of the 6-monthly scale. Hence, in general, the duration of the drought events recognized by 12-monthly scale is longer than that of the 6-monthly or 3-monthly scales.

In general, a drought is considered a long-term phenomenon, for the meteorological droughts persisting for less than three months do not normally lead to severe pernicious effects and therefore may not be regarded as persistent droughts. Due to paucity of space, the detailed assessment of drought events (derived using SPI-3, SPI-6, and SPI-12) having persisted over 3 months in Chandrapur district only are presented in Table 5. From Table 4 and Table 5, it can be noticed that there are 18 persistent (duration >3 months)

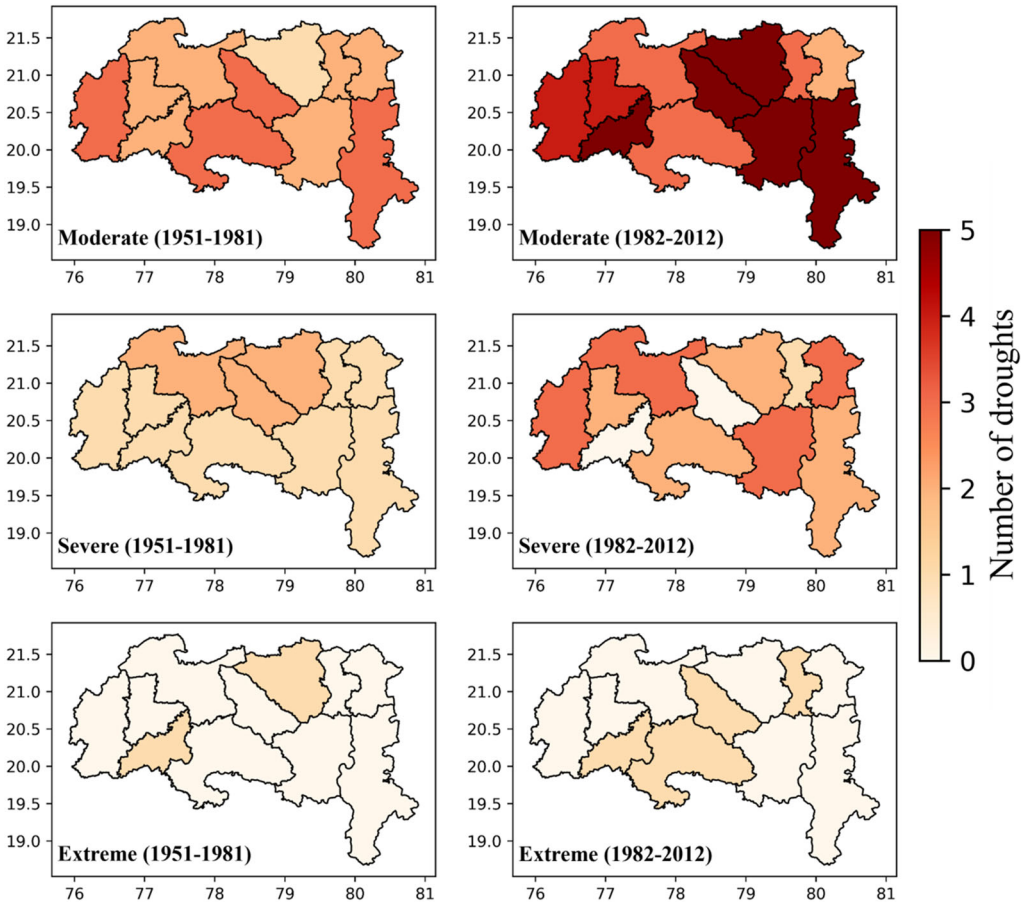
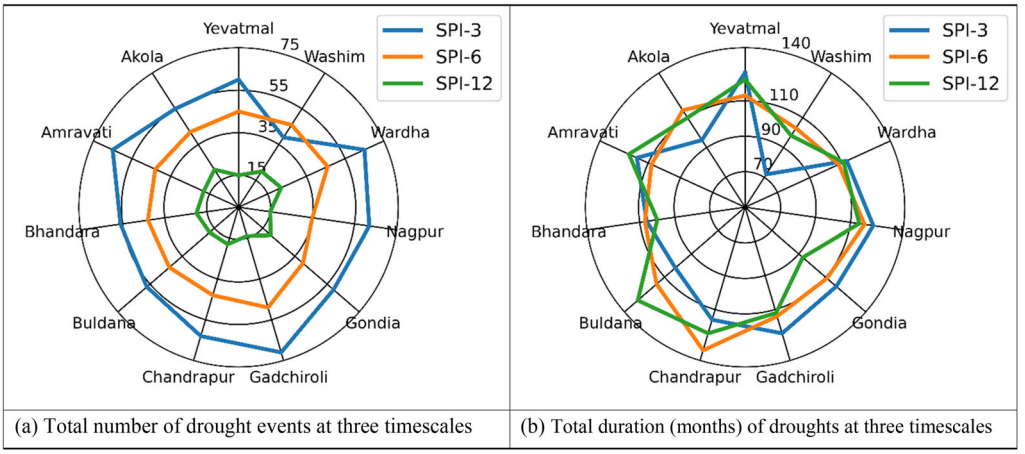


Figure 7. Number of droughts in Vidarbha division in 1951–1981 and 1982–2012 in each severity class.

drought events (out of 63) in the district when SPI-3 is used. Similarly, for SPI-6 and SPI-12, there are 15 (out of 43) persistent droughts and 11 (out of 18) persistent droughts, respectively, with varying intensities in Chandrapur district. It can be noticed that the number of drought events varies inversely with the length of time scale used. Therefore, the SPI-3-based drought events continue for very short durations, whereas these continue for longer duration but with reduced number of drought events at an increased time scale. The inter-arrival time of droughts, i.e., the time difference between two consecutive events can be estimated reliably at longer scales. For example, the year 1972 witnessed one of worst drought year over the state of Maharashtra. there was only a drought event continuing from June 1972 to July 1973 over the Chandrapur district, whereas there were three drought events during January 1972 to June 1973. Hence, SPIs of large time scales better capture drought persistence than those of short time scales.

Further, the number of persistent droughts during 1951–1981 and 1982–2012 as, for example, for Chandrapur district (Table 5) also signifies the changing meteorological conditions. It can be seen that there were 6, 5, and 3 persistent droughts based on SPI-3, SPI-6, and SPI-12, respectively, for the former epoch. These however increased to 12, 10, and 8, respectively, for the latter epoch. Hence, the frequency, duration, and severity of droughts have increased during 1982–2012.



(a) Total number of drought events at three timescales | (b) Total duration (months) of droughts at three timescales

Figure 8. Number of drought events and total drought duration for the districts under different timescales.

Table 4. Variation of SPI-based drought characteristics with varying timescales in all districts.

District	Timescale (months)	Drought duration (months)	No. of drought events	Average drought intensity
Yevatmal	3	126	60	−1.364
	6	113	45	−1.448
	12	122	15	−1.421
Washim	3	72	39	−1.406
	6	103	46	−1.449
	12	98	20	−1.497
Wardha	3	113	65	−1.389
	6	108	46	−1.448
	12	111	22	−1.404
Nagpur	3	123	62	−1.418
	6	118	35	−1.494
	12	115	15	−1.497
Gondia	3	118	59	−1.432
	6	111	40	−1.476
	12	93	20	−1.445
Gadchiroli	3	124	71	−1.366
	6	114	49	−1.415
	12	112	14	−1.425
Chandrapur	3	116	63	−1.369
	6	134	43	−1.360
	12	124	18	−1.405
Buldana	3	102	57	−1.408
	6	116	43	−1.463
	12	130	18	−1.453
Bhandara	3	106	56	−1.430
	6	107	43	−1.479
	12	100	20	−1.425
Amravati	3	117	65	−1.393
	6	108	43	−1.497
	12	122	18	−1.486
Akola	3	95	55	−1.437
	6	115	42	−1.439
	12	110	21	−1.400

The trend analysis of droughts for 1951–2012 is carried out by applying the non-parametric MMK test to SPI-3, SPI-6, and SPI-12 time series for eleven districts and the results are presented in Figure 9, whereas the exact Z_{MMK} values are provided in Table 6. It can be observed that the trend is not significant for SPI-3 and SPI-6 series at most

Table 5. Detailed assessment of SPI-based drought characteristics of Chandrapur district at 3-, 6-, and 12-month time scales.

SPI-3						SPI-6						SPI-12					
S. No.	Initiation	End	Duration	Severity	Intensity	S. No.	Initiation	End	Duration	Severity	Intensity	S. No.	Initiation	End	Duration	Severity	Intensity
1	July 1965	Oct 1965	4	−4.80	−1.200	1	Apr 1954	June 1954	3	−3.42	−1.140	1	Aug 1965	May 1966	10	−12.45	−1.245
2	Jan 1972	Mar 1972	3	−4.55	−1.517	2	July 1965	Jan 1966	7	−8.74	−1.249	2	June 1972	July 1973	14	−22.47	−1.605
3	July 1972	Oct 1972	4	−6.30	−1.575	3	July 1972	Feb 1973	8	−12.40	−1.550	3	Sept 1974	May 1975	9	−11.33	−1.259
4	Apr 1973	June 1973	3	−4.37	−1.457	4	June 1974	Dec 1974	7	−9.18	−1.311	4	Aug 1982	June 1983	11	−14.22	−1.293
5	June 1974	Sept 1974	4	−5.59	−1.398	5	Jan 1977	Apr 1977	4	−5.38	−1.345	5	Sept 1984	Sept 1985	13	−20.54	−1.580
6	Oct 1976	Dec 1976	3	−3.55	−1.183	6	July 1982	Nov 1982	5	−7.07	−1.414	6	Aug 1987	June 1988	11	−16.50	−1.500
7	June 1982	Sept 1982	4	−5.65	−1.413	7	June 1984	Feb 1985	9	−13.56	−1.507	7	Oct 1989	Feb 1990	5	−5.81	−1.162
8	June 1984	Nov 1984	6	−8.34	−1.390	8	June 1987	Feb 1988	9	−12.18	−1.353	8	Oct 1996	Aug 1997	11	−15.32	−1.393
9	June 1987	Oct 1987	5	−6.96	−1.392	9	Oct 1989	Dec 1989	3	−3.41	−1.137	9	Oct 2002	June 2003	9	−10.61	−1.179
10	Sept 1989	Nov 1989	3	−3.33	−1.110	10	Jan 1992	Apr 1992	4	−6.46	−1.615	10	Aug 2004	May 2005	10	−12.73	−1.273
11	Oct 1991	Dec 1991	3	−4.37	−1.457	11	Apr 1996	Dec 1996	9	−11.43	−1.270	11	June 2009	June 2010	13	−20.54	−1.580
12	Feb 1992	Apr 1992	3	−3.91	−1.303	12	Oct 2002	Dec 2002	3	−3.94	−1.313						
13	Jan 1996	Mar 1996	3	−3.30	−1.100	13	July 2004	Dec 2004	6	−8.98	−1.497						
14	June 1996	Sept 1996	4	−4.80	−1.200	14	Feb 2009	Feb 2010	13	−20.32	−1.563						
15	July 2004	Nov 2004	5	−6.62	−1.324	15	Mar 2012	June 2012	4	−6.41	−1.603						
16	Nov 2008	Apr 2009	6	−8.15	−1.358												
17	Aug 2009	Nov 2009	4	−6.05	−1.513												
18	Apr 2012	June 2012	3	−5.05	−1.683												

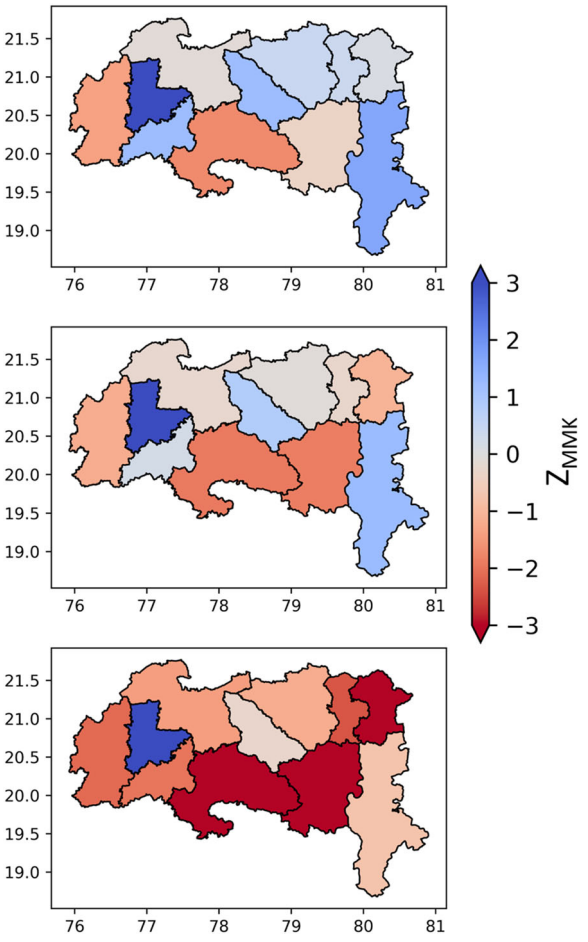


Figure 9. District-wise results of MMKT for trend analysis of SPI at different timescales.

Table 6. Z_{MMK} for SPI time series at different timescales over individual districts.

	Z_{MMK}		
Districts	SPI-3	SPI-6	SPI-12
Yevatmal	−1.688*	−1.899*	−3.691***
Washim	1.242	0.224	−1.978**
Wardha	1.242	0.835	−0.304
Nagpur	0.436	−0.073	−1.195
Gondia	0.099	−1.077	−3.134***
Gadchiroli	1.663*	1.254	−0.768
Chandrapur	−0.355	−1.860*	−5.063***
Buldana	−1.378	−1.185	−2.125**
Bhandara	0.376	−0.270	−2.351**
Amravati	−0.105	−0.239	−1.425
Akola	3.377***	4.119***	4.051***

Note: *, **, *** denote trends at 10%, 5%, and 1% significance levels, respectively.

stations whereas 10 out of 11 stations exhibit a significant negative trend at SPI-12 time series (Table 6). Here, it is notable that SPI being the reduced variate represents the rainfall amounts in a standardized form. Therefore, a decreasing trend of SPI time series shall represent a decreasing pattern of rainfall, indicating further worsening of meteorological

drought conditions. Table 6 shows that Z_{MMK} of the SPI series becomes more negative with increasing timescale, indicating the efficacy of SPI-12 to capture the drought trends, consistent with the increasing frequency of droughts in the recent epoch as discussed earlier.

It is interesting to note that Akola is the only district showing a positive trend of SPI and Z_{MMK} being significant at 1% significance level at all timescales. However, the number of droughts has doubled in 1982-2012 when compared with that of 1951–1981 (Table 3). The analysis thus requires a detailed investigation of SPI values, as follows. It is found that during 1951–1981, there were 3 dry years (SPI < -1) and only 2 wet years (SPI $>$ 1). Further, there were also 9 near-normal wet years, i.e., SPI based on annual rainfall varied from 0 to 1. However, during 1982–2012, the number of wet years and near-normal wet years increased to 5 and 12, respectively. Although this might have resulted in an increasing trend of SPI in the district, there is an increase in both dry and wet years and a decrease in a number of near-normal years ($-1 <$ SPI < 1). Tacitly, it is also an indication of climate change in the district and, in turn, increasing both the number of droughts and floods.

The differences in the Z_{MMK} for SPI series at different timescales is intriguing and urges for further investigations to unveil the best timescale to consider for drought trends. The frequent flickering or randomness in the time series is an important factor, which affects the statistical inferences of a time series. In a highly fluctuating series, the estimation of parameters becomes difficult (Wu et al. 2005). The randomness can be observed visually from the time series. The time series of SPI-3, SPI-6 and SPI-12 for each district is shown in Figure 10. It can be observed that the SPI-3 time series possesses more significant fluctuations or randomness for all districts relative to those corresponding to SPI-6 or SPI-12 series despite the fact that the number of data points for the SPI series at different timescales is the same. This indicates that the randomness decreases as the timescale increases, thereby being the lowest in the SPI-12 series for all districts. Moreover, the highly excessive or deficit rainfall in a particular month may lead to deceived notions at shorter timescales (Djellouli et al. 2016; Vicente-Serrano 2006; Wu et al. 2005). The number of tuning points for SPI at shorter scales is expected to be higher due to the frequent fluctuations. Because the trend analysis intends to determine the monotonicity in time series, a larger number of turning points may mislead the MMK test results. Considering all the above justifications, it can be inferred that the trends associated with SPI-12 is more reliable than SPI-3 or SPI-6.

Apart from visualization, the randomness present in a time series can be expressed in different ways. One of the simplest techniques used in literature is the 'moving average' method, which smoothens out the high-frequency fluctuations of a time series, thereby enabling the trend (if any) to be visible (Subramanya 2008). It is pertinent to mention that the moving average will be a smoother line if the fluctuations are lesser, and in such cases, the difference between the observation/data in a time series and the corresponding moving average is also relatively small. For the sake of consistency, a 5-month moving average is considered for SPI series at 3-, 6-, and 12-monthly scales for each district in Vidarbha Region. However, due to paucity of space, the moving average plots along with the SPI time series at different scales for only one district, i.e., Chandrapur district, are presented in Figure 11.

It can be noticed from Figure 11 that the 5-month moving average is having an excellent reconciliation with the SPI-12 series, thereby achieving the maximum smoothness among all timescales, whereas the contrasting case is observed for SPI-3. The mean, standard deviation and the squared difference from the moving average for each SPI time

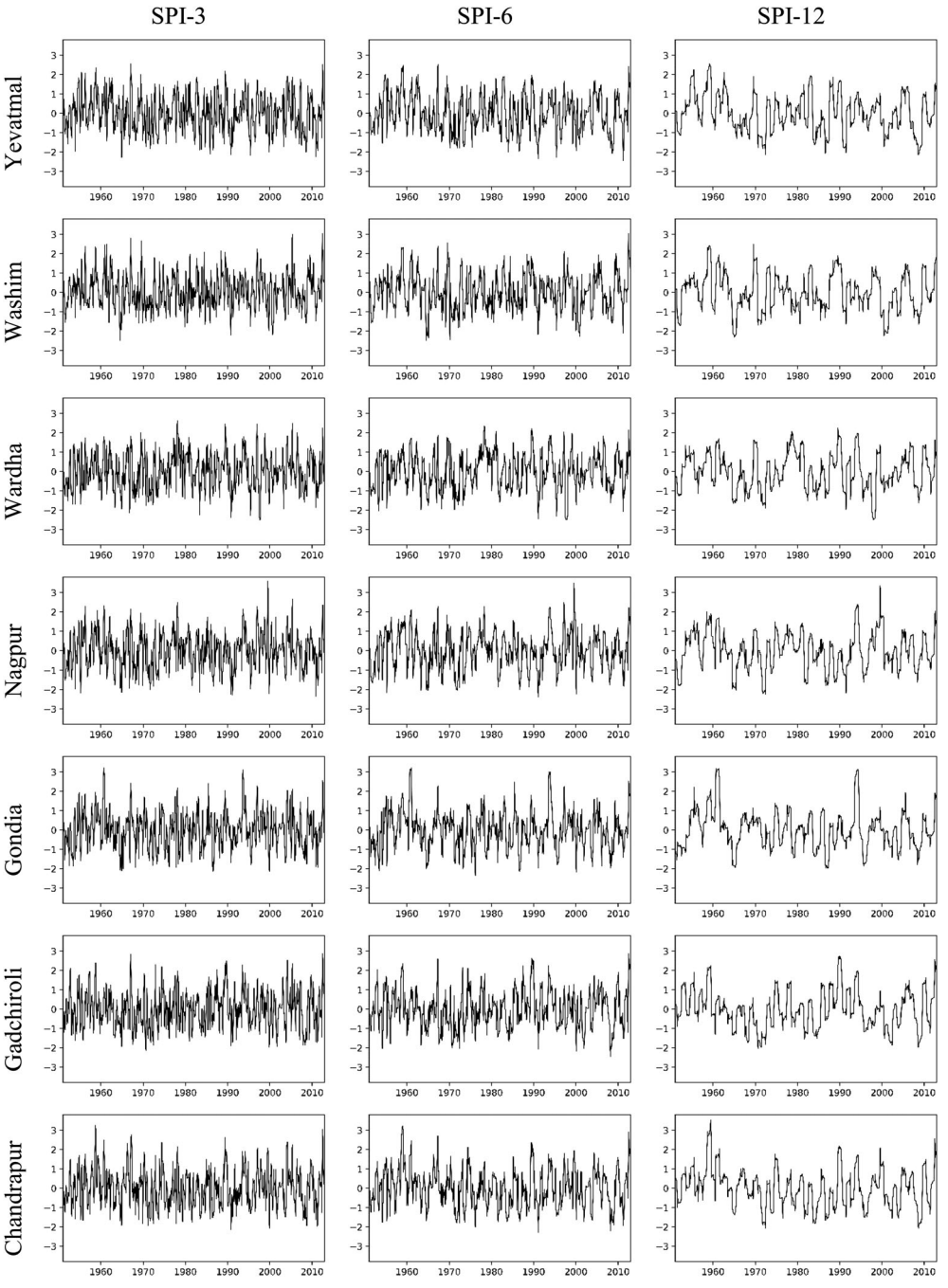


Figure 10. Time series of 3-, 6- and 12-month SPI for 1951–2012 over each constituent district of the Vidarbha Region.

series for the districts in the Vidarbha Region are presented in Table 7. SPI being a standard normal variate, the mean and the standard deviation should be ideally zero and one, respectively. It can be observed that the mean and the standard deviation of SPI-12 series is closer to the ideal values of zero and one, respectively, for almost all the districts when

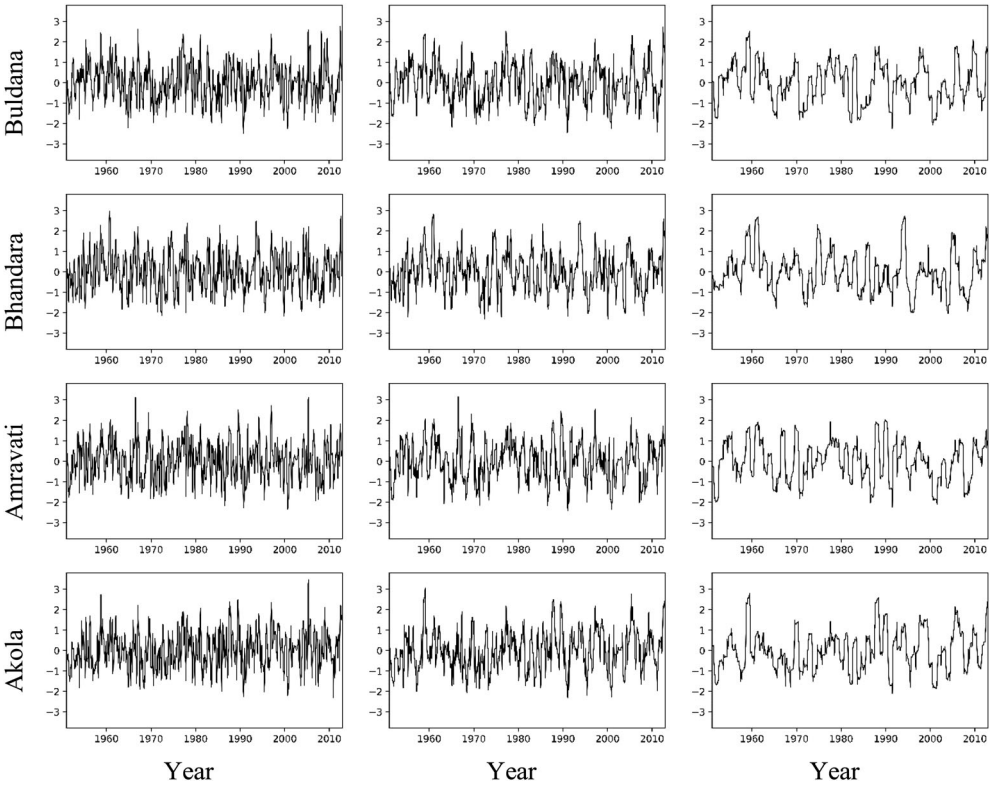


Figure 10. Continued.

compared to SPI-6 or SPI-3. The frequent flickering or randomness in the time series affects the statistical inferences of a time series, as discussed previously. However, the difference between the three SPI series in terms of mean or standard deviation is very small. Nevertheless, the difference between the SPI and the moving average time series at different timescales is significant. For all the districts, this difference ranges from 405.78 to 466.06, 268.04 to 306.57, and 116.32 to 138.68 for SPI-3, SPI-6, and SPI-12, respectively. This clearly justifies the higher randomness/fluctuations in SPI-3, which typically reduces with the increase in timescale.

The moving average is a crude technique to visualize the trend of any time series, and it is desirable to have the least difference between the data in a time series and its corresponding moving average value. This enables drawing inferences about a particular month's dryness/wetness condition from the moving average time series with higher confidence. The efficacy of SPI-12 to capture the drought trends is evident from its consistency with the epoch-wise analysis of increasing drought frequency in the recent period over almost all the districts. On the other hand, it is difficult to discern the SPI-3 or SPI-6 value of a particular month from the 5-month moving average. As mentioned previously, it may be attributed to the randomness in time series of SPI-3 or SPI-6 series, leading to failure in capturing definite trends.

Overall, this study provides key insights on the frequency of drought years over the Vidarbha Region. The increasing frequency and severity of drought years in the recent period will be a serious concern for the region, especially from the point of view of water resources planning and management. As the region is dominated by agricultural lands, the

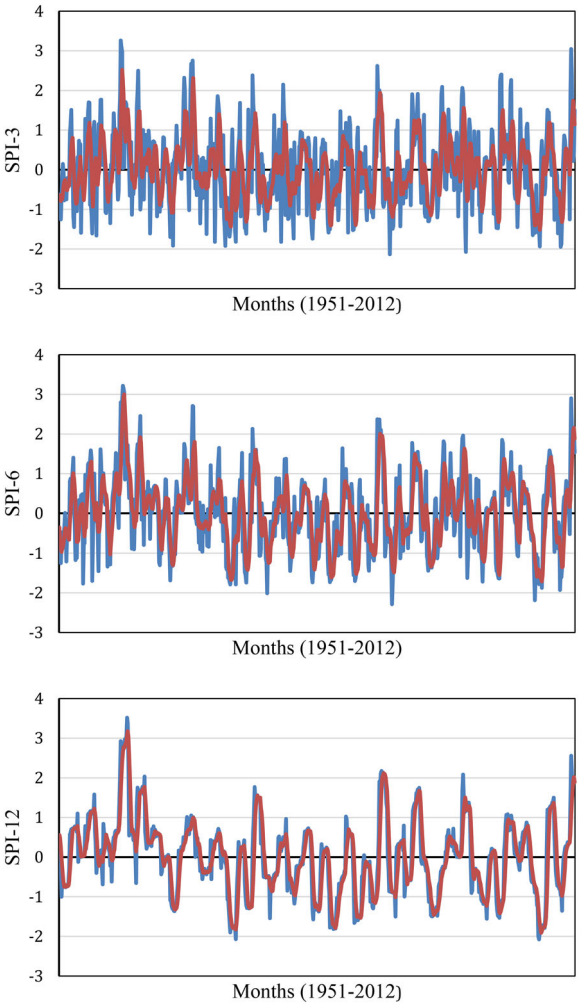


Figure 11. The time series of SPI and the corresponding 5-month moving average at different timescales for Chandrapur district. Note: The blue and the maroon lines represent SPI series and 5-months moving average, respectively.

Table 7. Mean, standard deviation and the squared difference from the moving average for each SPI time series for the distrcits in the Vidarbha Region.

	Mean			SD			Squared difference from the moving average time series		
Districts	SPI-3	SPI-6	SPI-12	SPI-3	SPI-6	SPI-12	SPI-3	SPI-6	SPI-12
Yevatmal	0.0278	0.0184	0.0154	0.9552	0.9690	0.9751	438.84	301.68	132.64
Washim	0.0587	0.0255	0.0194	0.9099	0.9527	0.9635	415.97	304.60	126.06
Wardha	0.0332	0.0279	0.0236	0.9364	0.9384	0.9467	405.78	268.04	125.16
Nagpur	0.0236	0.0183	0.0185	0.9622	0.9681	0.9673	442.51	283.55	123.25
Gondia	0.0255	0.0172	0.0128	0.9605	0.9709	0.9791	441.38	286.26	116.32
Gadchiroli	0.0207	0.0173	0.0134	0.9628	0.9707	0.9795	466.06	306.57	125.39
Chandrapur	0.0164	0.0132	0.0140	0.9694	0.9775	0.9781	429.34	281.58	134.06
Buldana	0.0323	0.0201	0.0169	0.9404	0.9630	0.9729	423.68	271.65	128.04
Bhandara	0.0272	0.0193	0.0131	0.9590	0.9679	0.9790	435.29	290.01	128.29
Amravati	0.0286	0.0212	0.0179	0.9487	0.9637	0.9703	438.05	274.38	138.68
Akola	0.0364	0.0159	0.0127	0.9418	0.9721	0.9805	434.26	271.56	135.43

results emphasize an immediate attention towards adaptive and mitigative measures to reduce the pernicious impacts of droughts. The dependence of timescales on drought characteristics are aptly discussed. The drought events continue for very short durations at shorter timescales, whereas these continue for longer duration but with reduced number of drought events at an increased time scale. Hence, the inter-arrival time of droughts can be estimated reliably at longer scales. The power of statistical tests for estimating trends in a SPI time series also reduces with decreasing timescale. The detailed analysis and information reported in this study provide useful insights to reckon the drought trends in a time series, particularly in multi-scalar assessments. However, this study is associated with some limitations. The constraints of data availability have led to analyze the drought characteristics only up to 2012. Although the duration of 62 years is reasonably long to identify the drought characteristics of the districts with substantial confidence, it ignores any drastic changes in the rainfall pattern that may have occurred in the recent years under the increasing influence of climate change. Besides, the present study has utilized the univariate SPI, which is computed using only rainfall data. However, temperature is also a crucial factor that may affect the climatic water availability. An investigation of drought characteristics using the multivariate indices, e.g., Standardized Precipitation Evapotranspiration Index (SPEI), Reconnaissance Drought Index (RDI), Standardized Precipitation Temperature Index (SPTI), etc., can be useful, especially using the recent years' data. Further, the sharply increasing frequency and severity of droughts in recent years also motivate to carry out drought assessments using non-stationarity considerations. These are the future scopes, which may contribute towards enhancing/improving the information reported in this study.

5. Conclusions

In this study, a detailed assessment of SPI-based meteorological drought characteristics in Vidarbha division/region, India, is carried out using the monthly (62 years, 1951–2012) rainfall data of eleven districts. From the analysis, the following conclusions are drawn:

1. The annual rainfall-based SPI results show that Vidarbha division is prone to droughts and usually undergoes droughts once in 5 to 8 years in its different districts.
2. There is a sharp increase in severity and frequency (once in 3 to 6 years) of droughts during 1982–2012 compared to those during 1951–1981.
3. The number of drought events reduces with increasing timescale. However, the persistence of droughts is better captured at longer timescales.
4. SPI-12 is better able to capture the definite trends than SPI-3 or SPI-6, largely due to higher fluctuations in SPI-3 or SPI-6 time series, affecting the Mann-Kendall test results.
5. The trends of SPI-12 in 10 out of 11 districts have been negative, but most being statistically significant, implying an increasing trend of drought conditions in Vidarbha division.

Disclosure statement

No potential conflict of interest was reported by the authors.

Data availability statement

The data that support the findings of this study are available from the authors upon reasonable request.

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